A Machine-Checked Obstruction to Hypothesis (H1) in the SG–Cole–Hopf Route to 3D Navier–Stokes Global Regularity

The MeTTapedia formalization effort

June 2026

Abstract

Ben Goertzel's manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program* (v7) reduces global regularity of the 3D incompressible Navier–Stokes equation on $\mathbb{T}^3$ to a single uniform-in-time identity-energy bound on the volume-preserving diffeomorphism group $\mathrm{SG} = \mathrm{SDiff}(\mathbb{T}^3)$. That bound is in turn obtained from two purely *local* hypotheses on a chart-cutoff region K, the first of which, hypothesis (H1), asks that the adjoint / push-forward operator Ad_g be uniformly bounded as a map $H^m \to H^m$ for all g in an H^m chart-neighborhood of the identity. We record a machine-checked finding of the MeTTapedia formalization effort: hypothesis (H1) is *false* in a clean Fourier-mode shear model. On the normalized two-torus we exhibit a one-parameter family of area-preserving shear displacements of *fixed* H^m size ε (hence inside any prescribed chart radius), whose adjoint operator ratio equals $\sqrt{1+(\varepsilon k)^2}$ and therefore exceeds every finite constant as the wavenumber $k \to \infty$. The statement and three supporting theorems are formalized with no `sorry` in the Lean module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`; an accompanying numeric lab *derives* the relevant Sobolev norms from the weight structure rather than assuming them, isolating the Ebin–Marsden derivative-loss mechanism (Ad_g loses one derivative: $\mathrm{Ad}_g v = (Dg)\,(v \circ g^{-1})$, and $g \in H^m \Rightarrow Dg \in H^{m-1}$). We emphasize the exact scope: this is a refutation *of (H1) within a model*, not a from-first-principles Sobolev theorem, and it says nothing about Navier–Stokes regularity itself, which remains open. The consequence for the route is precise and limited: the SG–Cole–Hopf program does not establish global regularity, because the single hypothesis on which it rests does not hold as stated.

1 The problem and the role of hypothesis (H1)

1.1 The SG–Cole–Hopf route

We work on the flat three-torus $\mathbb{T}^3$ with smooth divergence-free initial data u_0. Ben Goertzel's manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7 (hereafter "v7"), lifts the incompressible Navier–Stokes equation to a diffusion on the volume-preserving diffeomorphism group

$$\mathrm{SG} := \mathrm{SDiff}(\mathbb{T}^3),$$

equipped with a right-invariant Fourier frame $\{D_i\}_{i\geq 1}$ induced by a divergence-free frame $\{E_i\}_{i\geq 1}$ on $\mathbb{T}^3$. The generator and carré du champ are

$$L := \sum_{i\geq 1} D_i^2, \qquad \Gamma(F) := \sum_{i\geq 1} (D_i F)^2.$$

One solves the SG heat equation $\partial_t \Phi = \nu L \Phi$ with a strictly positive datum $\varphi = \exp(-W_0/(2\nu))$, applies the Cole–Hopf transform $W = -2\nu \log \Phi$ to obtain an SG Hamilton–Jacobi–Bellman potential, and pushes the Γ-gradient down at the identity to a projected Navier–Stokes solution $u(t)$ on $\mathbb{T}^3$.

The analytic content is concentrated entirely in closing the Beale–Kato–Majda (BKM) vorticity criterion. With $u(t) = \mathrm{Ev}(W(t))$ and $\omega(t) = \operatorname{curl} u(t)$, the push-down identity and Cauchy–Schwarz give a frame constant $C_{\mathrm{curl}E} < \infty$ with

$$\|\omega(t)\|_{L^\infty(\mathbb{T}^3)} \leq \sqrt{C_{\mathrm{curl}E}}\,\sqrt{\Gamma(W(t))(\mathrm{id})}, \tag{1}$$

so that a uniform-in-time bound on $\Gamma(W(t))(\mathrm{id})$ makes the BKM integral $\int_0^T \|\omega(t)\|_{L^\infty}\,dt$ finite on every finite interval. By the log chain rule $\Gamma(\log \Phi) = \Gamma(\Phi)/\Phi^2$ and the maximum principle $\Phi(t, \mathrm{id}) \geq m_\varphi$ (v7, Prop. 7.1),

$$\Gamma(W(t))(\mathrm{id}) = 4\nu^2\,\Gamma(\log \Phi(t))(\mathrm{id}) \leq \frac{4\nu^2}{m_\varphi^2}\,\Gamma(P_t\varphi)(\mathrm{id}), \tag{2}$$

which reduces the whole program to a single uniform bound on $\Gamma(P_t\varphi)(\mathrm{id})$.

1.2 Reduction to a local energy bound, and hypothesis (H1)

Rather than invoking a global Bakry–Émery Γ_2 curvature lower bound, v7 obtains that single bound from a *local* package. Let $\kappa : U \to V \subset H^m_\sigma(\mathbb{T}^3)$ be the chart of a small SDiff^m neighborhood of the identity ($m \geq 5$ integer, with the Γ-Hilbert structure at id equivalent to the H^m_σ norm), let χ be the cutoff defining W_0 and φ, and set the cutoff region

$$K := \kappa^{-1}(\operatorname{supp}\chi) \subset U.$$

Outside K one has $W_0 \equiv 0$, hence $\varphi \equiv 1$ and $\Gamma(\varphi) \equiv 0$. On K, v7 assumes two local hypotheses (v7, §5.3 and §6, displayed at Lemma D.18):

$$\text{(H1) Adjoint bounded on support:}\quad \exists\, C_{\mathrm{Ad}} < \infty : \ \|\mathrm{Ad}_g\|_{H^m \to H^m} \leq C_{\mathrm{Ad}} \quad \text{for all } g \in K, \tag{3}$$

$$\text{(H2) Finite cutoff energy:}\quad \exists\, C_\varphi < \infty : \ \Gamma(\varphi)(g) \leq C_\varphi^2 \quad \text{for all } g \in K. \tag{4}$$

From (3)–(4) v7 derives the key estimate via a conjugation differentiation identity (v7, Prop. 6.5, proved as Lemma F.7): for all $t \geq 0$,

$$\Gamma(P_t\varphi)(\mathrm{id}) \ \leq\ C_{\mathrm{Ad}}^2\, C_\varphi^2. \tag{5}$$

The mechanism is transparent. In the flow representation $P_tF(g) = \mathbb{E}[F(G_t \circ g)]$, differentiating along a conjugation curve $C_{t,i}(\varepsilon) = g \circ \exp(\varepsilon E_i) \circ g^{-1}$ produces the direction $\mathrm{Ad}_g E_i$, and Cauchy–Schwarz in the Γ-Hilbert structure gives the pointwise bound

$$\left|D_{\mathrm{Ad}_g E_i}\varphi(g)\right|^2 \leq \|\mathrm{Ad}_g\|_{\mathrm{op}}^2\,\Gamma(\varphi)(g) \leq C_{\mathrm{Ad}}^2\, C_\varphi^2\, \mathbf{1}_{\{g \in K\}}, \tag{6}$$

after which a Jensen/energy reduction yields (5). Chaining (5) through (2) and (1) closes BKM. Hypothesis (H1) is therefore the single load-bearing analytic input of the route: *every* subsequent estimate inherits the constant C_{Ad} from (3), and the program establishes global regularity if and only if (3) holds.

1.3 How v7 attempts to verify (H1)

v7 verifies (H1) at Lemma 6.1 (and again at Lemma D.18). The adjoint / push-forward action is written, for $g \in \mathrm{SDiff}^m$ and $v \in H^m_\sigma$, as

$$\mathrm{Ad}_g v := (Dg)\,(v \circ g^{-1}), \tag{7}$$

and the argument is that, for $m > 5/2$, composition and inversion are continuous (indeed smooth) operations on SDiff^m, so the map $(g, v) \mapsto \mathrm{Ad}_g v$ is continuous from $\mathrm{SDiff}^m \times H^m_\sigma$ into H^m_σ; hence $g \mapsto \|\mathrm{Ad}_g\|_{\mathrm{op};\, H^m \to H^m}$ is locally bounded near id, and one shrinks K until (3) holds.

The gap is in (7) itself. The factor Dg is the Jacobian of g, and a map $g \in H^m$ has Jacobian $Dg \in H^{m-1}$ only. The adjoint thus carries a *derivative loss*: Ad_g maps H^m into H^{m-1}, and there is no uniform $H^m \to H^m$ operator bound near the identity in the H^m topology. This is the classical Ebin–Marsden phenomenon: right translation on SDiff^m is smooth, but the adjoint of the right-invariant frame is not a bounded H^m operator uniformly over an H^m-neighborhood. The local-boundedness argument conflates topological-group continuity of composition with the operator-norm estimate (3) that the route actually needs.

2 The Fourier-mode shear model and the checked ¬(H1) theorems

We make the derivative-loss obstruction precise and machine-checkable in a clean analytic model, on the normalized two-torus, of an area-preserving shear family

$$g_k(x, y) = \big(x,\ y + a_k \sin(kx)\big), \qquad \delta_k := g_k - \mathrm{id} = \big(0,\ a_k \sin(kx)\big).$$

The amplitude a_k is chosen so that the displacement has *fixed* H^m *size* ε for every wavenumber k; thus the whole family sits inside any prescribed chart radius. Applying the push-forward (7) to the constant horizontal field $v = (1, 0)$ gives

$$\mathrm{Ad}_{g_k} v = \big(1,\ a_k\, k\, \cos(kx)\big),$$

and, with the same H^m normalization, the operator ratio is

$$\frac{\|\mathrm{Ad}_{g_k} v\|_{H^m}}{\|v\|_{H^m}} = \sqrt{1 + (\varepsilon k)^2}.$$

The displacement stays at size ε while the ratio grows without bound: no finite C_{Ad} can dominate it on any positive chart radius.

This is formalized, with no `sorry` and by elementary means, in the Lean module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`. The model quantities are defined as follows.

```
/-- In the normalized two-torus shear family, every selected displacement has
fixed ‘H^m‘ size ‘epsilon‘. -/
def benH1ModeDisplacementHNorm (epsilon : Real) (_k : Nat) : Real :=
  epsilon

/-- Lower-bound ratio obtained by applying the shear push-forward to the
constant horizontal vector field. -/
def benH1ModeAdjointRatio (epsilon : Real) (k : Nat) : Real :=
  Real.sqrt (1 + (epsilon * (k : Real)) ^ 2)
```

```
/-- The linearized commutator part alone grows like ‘epsilon * k‘. -/
def benH1ModeCommutatorRatio (epsilon : Real) (k : Nat) : Real :=
  epsilon * (k : Real)
```

Hypothesis (H1), restricted to this normalized family inside a chart radius ρ (taking $\varepsilon = \rho/2$ so the family is strictly inside the ball), is the predicate:

```
/-- H1 restricted to the normalized Fourier-mode shear family inside chart
radius ‘rho‘. -/
def benH1FourierModeUniformBoundWithinRadius (rho C : Real) : Prop :=
  forall k : Nat,
    benH1ModeDisplacementHNorm (rho / 2) k <= rho ->
      benH1ModeAdjointRatio (rho / 2) k <= C
```

The core fact is that the adjoint ratio dominates the linear commutator part εk, which is unbounded over k. The proof is fully elementary: it picks $k > C/\varepsilon$, lower-bounds $\sqrt{1 + (\varepsilon k)^2}$ by εk, and derives a contradiction.

```
theorem benH1ModeAdjointRatio_unbounded {epsilon C : Real}
    (hepsilon : 0 < epsilon) :
    Not (forall k : Nat, benH1ModeAdjointRatio epsilon k <= C) := by
  intro hbound
  obtain ⟨k, hk⟩ := exists_nat_gt (C / epsilon)
  have hC_lt : C < epsilon * (k : Real) := by
    have hmul := mul_lt_mul_of_pos_left hk hepsilon
    field_simp [ne_of_gt hepsilon] at hmul
    exact hmul
  have hnonneg : 0 <= epsilon * (k : Real) := by positivity
  have hlinear_le_ratio :
      epsilon * (k : Real) <= benH1ModeAdjointRatio epsilon k := by
    unfold benH1ModeAdjointRatio
    have hsquare :
        (epsilon * (k : Real)) ^ 2 <= 1 + (epsilon * (k : Real)) ^ 2 := by
      linarith
    calc
      epsilon * (k : Real)
          = Real.sqrt ((epsilon * (k : Real)) ^ 2) := by
              rw [Real.sqrt_sq_eq_abs, abs_of_nonneg hnonneg]
      _ <= Real.sqrt (1 + (epsilon * (k : Real)) ^ 2) :=
              Real.sqrt_le_sqrt hsquare
  exact not_lt_of_ge (le_trans hlinear_le_ratio (hbound k)) hC_lt
```

From this the refutation of (H1) on the model follows directly. Any positive chart radius ρ admits the normalized family at $\varepsilon = \rho/2$ (whose displacements all have H^m size $\rho/2 \le \rho$), and its adjoint ratios are unbounded, contradicting any proposed finite C.

```
/-- Any positive chart radius contains the normalized shear family with
‘epsilon = rho / 2‘, and its adjoint ratios are unbounded. -/
theorem not_benH1FourierModeUniformBoundWithinRadius {rho C : Real}
    (hrho : 0 < rho) :
    Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro hbound
  have hepsilon : 0 < rho / 2 := by positivity
```

```
  have hmode_bound : forall k : Nat, benH1ModeAdjointRatio (rho / 2) k <= C := by
    intro k
    apply hbound k
    unfold benH1ModeDisplacementHNorm
    linarith
  exact benH1ModeAdjointRatio_unbounded hepsilon hmode_bound

/-- Compact public verdict: Ben's H1, as a uniform `H^m -> H^m` adjoint bound,
is false on the normalized Fourier-mode shear model for every positive chart
radius and every proposed finite constant. -/
theorem benGoertzelH1_false_in_fourierModeShearModel :
    forall rho C : Real,
      0 < rho -> Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro rho C hrho
  exact not_benH1FourierModeUniformBoundWithinRadius (rho := rho) (C := C) hrho
```

The three checked statements are `benH1ModeAdjointRatio_unbounded`, `not_benH1FourierModeUniformBoundWithinRadius`, and the compact verdict `benGoertzelH1_false_in_fourierModeShearModel`: for every positive chart radius and every proposed finite constant, the modeled (H1) uniform $H^m \to H^m$ adjoint bound fails. (A companion lemma `benH1ModeCommutatorRatio_unbounded` records that the linearized commutator part εk is already unbounded on its own.)

3 The numeric lab: the norms are derived, not assumed

The Lean development encodes the two normalized quantities $\|\delta_k\|_{H^m} = \varepsilon$ and $\|\mathrm{Ad}_{g_k} v\|_{H^m}/\|v\|_{H^m} = \sqrt{1+(\varepsilon k)^2}$ as definitions. To confirm that these are not circular — that the factor k in the adjoint ratio is a genuine consequence of the Sobolev-weight structure and the derivative in (7), rather than an assumption — a companion numeric lab (`ns_ben_h1_fourier_mode_lab.py`, in the MeTTapedia validation lab) *computes* the norms from first quantities. With H^m Fourier weight $w_k = (1+k^2)^{m/2}$:

- the amplitude is set to $a_k = \sqrt{2}\,\varepsilon/w_k$, so the displacement norm is $\|\delta_k\|_{H^m} = |a_k|\,w_k/\sqrt{2} = \varepsilon$ (the $\sqrt{2}$ is the L^2 normalization of $\sin(kx)$);
- differentiating, $\partial_x\big(a_k \sin(kx)\big) = a_k k \cos(kx)$, whose H^m norm is $|a_k|\,k\,w_k/\sqrt{2} = \varepsilon k$. The weight w_k *cancels* the amplitude, leaving ε; the surviving factor k is the one derivative gained by Dg_k;
- the constant field has $\|v\|_{H^m} = 1$, so the adjoint norm is $\sqrt{1^2+(\varepsilon k)^2}$ and the ratio is $\sqrt{1+(\varepsilon k)^2}$.

This is exactly the Ebin–Marsden derivative loss of (7) made quantitative: $g_k \in H^m$ has $Dg_k \in H^{m-1}$, and the action of Dg_k multiplies the mode's high-frequency content by its wavenumber. Table 1 lists the lab output for $\varepsilon = 0.01$, $m = 3$. The displacement norm stays fixed at ε across four decades of k while the adjoint ratio grows.

4 Status and scope

We state the scope of this finding exactly.

k	amplitude a_k	$\|\delta_k\|_{H^m}$	commutator εk	ratio $\sqrt{1+(\varepsilon k)^2}$
8	2.699×10^{-5}	0.0100	0.080	1.0032
16	3.433×10^{-6}	0.0100	0.160	1.0127
32	4.310×10^{-7}	0.0100	0.320	1.0500
64	5.393×10^{-8}	0.0100	0.640	1.1873
128	6.743×10^{-9}	0.0100	1.280	1.6243
256	8.429×10^{-10}	0.0100	2.560	2.7484
512	1.054×10^{-10}	0.0100	5.120	5.2167
1024	1.317×10^{-11}	0.0100	10.240	10.2887

Table 1: Fourier-mode lab output ($\varepsilon = 0.01$, $m = 3$). The displacement H^m norm is pinned at ε for every wavenumber, while the adjoint operator ratio $\sqrt{1+(\varepsilon k)^2}$ grows without bound. For any proposed constant C the ratio first exceeds it near $k \approx \sqrt{C^2-1}/\varepsilon$ (e.g. $C = 2$ at $k = 174$, $C = 10$ at $k = 995$, $C = 100$ at $k = 10000$): no finite C_{Ad} survives.

What is established. Hypothesis (H1) of v7 — a uniform $H^m \to H^m$ operator bound (3) on the adjoint / push-forward Ad_g over an H^m chart-neighborhood of the identity — is *false* in the normalized Fourier-mode shear model of Section 2. The refutation is machine-checked (no `sorry`, elementary) in `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`, and the underlying norms are independently derived from the Sobolev-weight structure in the companion numeric lab. The model faithfully captures the mechanism that defeats v7's local-boundedness argument: the adjoint (7) loses one derivative ($g \in H^m \Rightarrow Dg \in H^{m-1}$), so on a fixed-radius H^m chart the operator norm is unbounded along high-wavenumber shears.

What is not established. This is a refutation of (H1) *in a model* — a single analytic Fourier-mode family together with the linearized commutator action — not a from-Mathlib-Sobolev-first-principles theorem about $\mathrm{SDiff}^m(\mathbb{T}^3)$. The model uses the exact push-forward formula (7) on an explicit shear family and the standard Fourier H^m norm; it does not formalize the full diffeomorphism-group functional analysis. The mechanism (derivative loss) is captured faithfully, but the strength of the conclusion is "checked in this model," not "checked over the entire chart from Mathlib's Sobolev theory."

Consequence for the route. Because every quantitative step of the SG–Cole–Hopf program inherits the constant C_{Ad} from (3) — through (5), (2), and (1) — the falsity of (H1) as stated means the route does *not*, as written, establish global regularity. v7 reduces 3D Navier–Stokes global regularity to (H1) (equivalently, to a uniform curvature / Bakry–Émery-type bound), and that reduction target does not hold uniformly in the form required.

What remains open. We make no claim about Navier–Stokes itself. Global regularity of the 3D incompressible Navier–Stokes equation remains an open problem; the finding here concerns only the viability of one particular proof route. Whether a repaired hypothesis — for instance an $H^m \to H^{m-1}$ adjoint bound combined with a separate mechanism to recover the lost derivative — could resurrect a version of the argument is left as a question for future work.

Provenance

The manuscript discussed is Ben Goertzel, *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7. The formalization, Fourier-mode model, and numeric lab are work of the MeTTapedia formalization effort; the Lean development is the module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`.